



Problem

The 3N+1 sequence is generated using the following rules: If n is even, the next number is $n / 2$. If n is odd, the next number is $3 \times n + 1$. The sequence continues until it reaches 1. The cycle length of a number is the total number of terms in the sequence, including the starting number and 1.

Given two integers i and j, determine the maximum cycle length among all integers between i and j (inclusive).

If $i > j$, swap them before processing.

Input Format

A single line containing two space-separated integers:

i j

Constraints

$1 \leq i, j \leq 1,000,000$

Output Format

Print three integers:

i j maximum_cycle_length

using the original values of i and j.

Sample Input 0

1 10

Sample Output 0

1 10 20

[Change Theme](#)

Language

Python 3



```
1 # Enter your code here. Read input from STDIN.
2 def cycle_length(n):
3     count=1
4     while n!=1:
5         if n%2==0:
6             n=n//2
7         else:
8             n=3*n+1
9         count+=1
10    return count
11 i,j=list(map(int, input().split()))
12 org_i,org_j=i,j
13 if i>j:
14     i,j=j,i
15 max_cycle=0
16 for num in range(i,j+1):
17     max_cycle=max(max_cycle,cycle_length(num))
18 print(org_i,org_j,max_cycle)
```

Line: 3 Col: 12



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0



Test case 1



Test case 2



Compiler Message

Success

